

4	(a) (i)	product of speed and density	M1	
		reference to speed in medium (and density of medium)	A1	[2]
	(ii)	α : ratio of reflected <u>intensity</u> and/to incident <u>intensity</u>	B1	
		Z_1 and Z_2 : (specific) acoustic impedances of media (on each side of boundary)	B1	[2]
	(b)	in muscle: $I_M = I_0 e^{-\mu x}$ $= I_0 \exp(-23 \times 3.4 \times 10^{-2})$	C1	
		$I_M / I_0 = 0.457$	C1	
		at boundary: $\alpha = (6.3 - 1.7)^2 / (6.3 + 1.7)^2$ $= 0.33$	C1	
		$I_T / I_M = [(1 - \alpha) =] 0.67$	C1	
		$I_T / I_0 = 0.457 \times 0.67$ $= 0.31$	A1	[5]

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